Upamana pramana

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Upamana pramana is acquiring knowledge with some analogous examples or similar known descriptions. It is the source of our knowledge regarding the relationship between names and things to be named. Upamana is assimilative knowledge. The base of knowledge is the resemblance or similarities and dissimilarities between two objects and their comparison to obtaining knowledge about unknown phenomena. Sushruta explained this as a separate tool for learning. [Su. Sa. Sutra Sthana. 1/16][1] Other ancient scholars like Vaisheshika, Sankhya, Yoga, Jain, Bauddha included this pramana under anumana pramana. It has immense utility in the field of medicine. The present article describes applications of analogy in nomenclature, diagnosis, and treatment of diseases.

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Etymology and derivation

The word 'upamana' is derived from Sanskrit verb 'ma', meaning to measure or to know and a prefix 'upa' meaning towards or near to or resemblance.[2]

Monnier Williams dictionary defines upamana as a comparison, resemblance, analogy, simile, or an object with which anything is compared.[3] Apte Sanskrit dictionary describes upamana as resemblance, comparison.[4]

Definition

In Charak Samhita, upamana pramana is described 'aupamya' in context of logical terms for reasoning (vaadamarga). It is the description of the similarity between two objects. Analogy of the disease dandaka (disease or condition where body becomes rigid like a rod) and danda (rod) is described. Dhanustambha (tetany oropisthotonos) is named due to its similarity with dhanushya (bow) and a physician with that of an archer. [Cha.Sa.Vimana Sthana 8/42]

Upamana is method of obtaining knowledge through the comparison with well-known or established fact. It is particularly instrumental in obtaining knowledge of objects that are minute, rejected, distant or complex to comprehend. [Dalhana on Su.Sa.Sutra Sthana.1/16].[1]

Tarkasangraha defines upamana as an instrumental cause or act of comparing two phenomena with each other. [Tarkasangraha 56][5] Finding the relation or comparison between the characteristic feature and one possessing it, is termed as 'upamiti'. It is the cognition through the means of upamana (analogy). Tarkasangraha gives an example of wild animal termed 'gavaya' which resembles a cow. When an individual is unaware about the identification or appearance of gavaya, on noticing the unknown animal he recollects the information regarding the resemblance with a cow and can predict that the animal is gavaya. Thus designation of gavaya is only possible after knowing its resemblance with a known entities like cow. This is possible with the help of upamana pramana.[6]

Synonyms

1. Sadrishya dnyana [Tarkasangraha]
2. Aupamya [Cha.Sa.Vimana Sthana.8/42]
3. Upamiti [Tarkasangraha]
4. Drishtanta [Cha.Sa.Vimana Sthana.8/31]
5. Udaharana [Cha.Sa.Vimana Sthana.8/149] , [Chakrapani on Cha.Sa.Vimana Sthana.8/3]

Types of upamana pramana

Nyaya philosophy describes three classifications of upamana pramana as below:

1. Sadrishya-vishishtapindadnyana (similarity-based)
2. Vaidharmya-vishishtapindadnyana (dissimilarity based)
3. Asadharanadharma vishishta (peculiarity based)

[Nyayavarthika Teeka on Nyaya Darshana][7]

Some scholars accept only two types of upamana i.e. one based on similarity (sadharmya-upamana) and one based on dissimilarity (vaidharmya-upamana).[8]

1. Similarity (sadrishya): The knowledge occurs based on the similarity between two objects or phenomena. For instance, the similarity of appearance between mole (tilakalaka) and sesame seed (tila).

2. Dissimilarity (vaidharmya): The knowledge is based on the dissimilarity in the comparable objects. For instance, a horse differs from an elephant due to lack of trunk. This enables observer to differentiate between the two. Similarly, the disease vomiting (chhardi) varies from loose stools (atisara) based on distinct prodromal signs.

3. Peculiarity (asadharana dharma vishishtapindadnyana): It is the peculiarity or specialty of a particular object or phenomenon that differentiates it from the rest of phenomena. For instance, the fever (jwara) is different from every other disease due to its peculiar symptom of pyrexia or santapa.

Importance

Upamana pramana helps to understand new, unknown or invisible objects on the basis of recognition and proven phenomena. This helps in generalization of concept that can be understood by individuals of varied intellect. It also helps in easy comprehension of the subject enabling the lay persons or beginners to understand the same. It is also important source of knowledge for other pramana like pratyaksha pramana, anumana pramana and aptopadesha. According to some theories, upamana pramana requires the knowledge of other pramana including direct perception (pratyaksha), inferential knowledge (anumana) and authoritative statements (aaptopadesha).

Importance in understanding Ayurveda physiology

The theory of similarity in man and the universe (Loka-Purusha Samya) is largely based on the upamana pramana. The purusha or individual is considered as epitome or miniature of the universe.[Cha.Sa.Sharira Sthana 5/5] The different organs and entities in the human body

represent the entities in the universe.[Cha.Sa.Sharira Sthana 4/13] For instance, in a person, prithvi, apa, tejas, vayu, akasha, and brahman are represented in structural form, moisture, heat, vital breath, space, and oneself (atman), respectively. Similarly, Indra in the universe is represented by ego in the person. Aditya (the sun) is represented in the catabolic processes. Rudra is manifestation of agitation, Soma (the moon) is cheerfulness, Vasu is happiness, Marut (the air) is enthusiasm, Ashwin is lustre, Vishvadeva are sense organs and their objects, darkness is ignorance, light is knowledge.[9] This relationship between microcosm and macrocosm is best determined by upamana pramana.

The different theories of transformation and nourishment of tissues (dhatuposhana nyaya) namely ksheeradadhi nyaya, kedarikulya nyaya, khalekapot nyaya are also based on the concept of upamana pramana.[Chakrapani commentary on Cha.Sa.Sutra Sthana.28/4]

The physiology in Ayurveda is largely based on functional aspects than anatomical or structural components. Thus, the functions of entities like dosha, dhatu, srotas can be understood using the similes of environmental components depicting resemblance in functions and appearance. The tridosha and their functioning is commonly compared with wind (vayu), sun (surya) and moon (soma), highlighting the application of upamana pramana.[Cha.Sa.Chikitsa Sthana.26/292] [Cha.Sa.Chikitsa Sthana.28/246]

The different maxims (nyaya) mentioned in Ayurveda like Shabdarchijalasantana nyaya also depict the similarities between the universe and individual physiology and guide the reader in understanding the basic concepts like digestion according to Ayurveda. [Su.Sa.Sutra Sthana.14/16][1] Different physiological concepts and assessment criteria like phenotype (prakriti), similarity of the phenotype of humans and animals (anukatva) [Su.Sa.Shareera Sthana.4/64-73][1] are explained with the help of upamana pramana.

Importance in understanding Ayurveda anatomy

Upamana pramana is most extensively used to describe the anatomical structures. The qualities and morphology of entities like semen (shukra), menstrual fluid (artava), bones (asthi), joints (sandhi), tendons (snayu), organs like heart (hridaya), bladder and urinary system (basti), body membranes (kala) are best described using the references from upamana pramana. Sushruta Samhita has thus extensively described various upamana for the above-mentioned entities in sharira sthana of the text. For instance, the functioning of cartilage and ligaments (snayu) is compared to that of a boat. [Su.Sa.Sharira Sthana.5/33][1] The appearance of female genitalia (yoni) is compared with conch (Shankha Nabhi). [Su.Sa.Sharira Sthana.5/43][1] Morphology of heart (hridaya) is comparable with lotus (pundrika).[Su.Sa.Sharira Sthana.4/31][1]

Importance in diagnosis

Upamana pramana has immense applicability in diagnosing different diseases. The basis of nomenclature of diseases is mainly dependent on upamana pramana. The table given below throws light upon the various names of diseases (vyadhi) derived by the application of upamana pramana.

Table 1: Nomenclature of diseases based on analogy

Name of the disease	The simile used for description	The rationale behind the description of simile
Dandalasaka	danda or rod	The body becomes stiff like a rod due to accumulation of ama in the body. [Cha.Sa.Vimana Sthana.8/42]
Vidari	vidari stem or tuber	The shape of papules/pustules/carbuncles (pidaka) is similar to Pueraria tuberosa or Indian Kudgu (vidari) stem.[Cha.Sa.Sutra Sthana 17/83]
Kacchapika	shell of tortoise	The shape of pidaka is hard and smooth like the shell of tortoise [Cha.Sa.Sutra Sthana 17/83]
Sarshapi	mustard seeds	Boils are similar to appearance of mustard seeds.[Cha.Sa.Sutra Sthana 17/83]
Madhumeha	honey	The appearance of urine is compared to that of honey. [Cha.Sa.Sutra Sthana 17/80]
Tilakalaka	sesame seeds	The appearance of pigmentation or mole is comparable to sesame (tila). [Cha.Sa.Sutra Sthana 18/25]
Alaji	agni (fire)	Appears and burns like fire. [Cha.Sa.Sutra Sthana 17/83]
Dhanustambha	dhanusha/bow	There is backward bending of body resembling bow like in opisthotonos. [Cha.Sa.Chikitsa Sthana 28/46]
Masurika	masura (red gram)	The appearance of pidaka is similar to that of red gram [Cha.Sa.Chikitsa Sthana 12/93]
Suryavarta	movement of the Sun	The manifestation/occurrence of pain in this condition is dependent on the time of day and hence the movement of the sun i.e. the pain in head increases with the rising of the sun. [Cha.Sa.Siddhi Sthana 9/79-83]

It is also used to decipher the symptoms of different diseases mentioned in the texts.

Table 2: Clinical features based on analogy

Disease	Upama rendered with meaning	Reference in the text
Kaphaja Hridriga	Ashmavatapeeda (feeling of heaviness in the chest as if a stone is placed over the chest)	[Cha.Sa.Sutra Sthana 17/35]
Krimija Hridroga	Suchivatpeeda (pricking pain like the needle is pricked onto the skin) Shastra-chhedanavatpeeda (pain similar to injury by a weapon)	[Cha.Sa.Sutra Sthana 17/39]
Vidradhi	Chronic vidradhi is compared to shastra, vidyut and agni i.e., weapon, lightening and fire, depicting its grave consequences like injury to vital organs (marmachhedana), fatality	[Cha.Sa.Sutra Sthana 17/103]
Sanyasa	Kashthibhootamritopama (rigid like a rod) depicts that the unconscious person appears to be in the vegetative stage.	[Cha.Sa.Sutra Sthana 24/44]
Pittaja raktapitta	The different colours are used to depict the appearance of raktapitta, Gomutrasannibha i.e. similar to cow's urine, mechakagaradhoomabha- similar to smoke, anjanabh (appears like collyrium)	[Cha.Sa.Chikitsa Sthana 4/12]
Pittaja Gulma	Vranavat means the pain and burning sensation in pittaja gulma is compared with that of injury or wound	[Cha.Sa. Chikitsa Sthana 5/13]
Tridoshaja Gulma	Ashmavat means the tenderness or sensation of tridoshaja gulma is similar to a stone	[Cha.Sa.Chikitsa Sthana 5/17]
Kaksha	Yajnopaveetapratima compared to Hindu sacred thread worn by males as it resembles the lesion that occurs in the part of the axilla.	[Cha.Sa.Chikitsa Sthana 12/91]
Pleehodara	Kachapasansthan - shell of tortoise - The texture of pleehodara is hard and smooth, like the tortoise shell.	[Cha.Sa.Chikitsa Sthana 13/37]
Udakodara	Udakapoornadutikshobha (leather bag filled with water) on the palpation of abdomen (udara), in ascites it feels like a bag filled with water.	[Cha.Sa.Chikitsa Sthana 13/47]
Appearance of arsha	Appearance of different types of arsha (haemorrhoids) using the examples of commonly observed phenomena like yakitakhandopama (similar to lobe of liver) etc.	[Cha.Sa.Chikitsa Sthana 14]
Kamala	Bhekavarna (toad like) appearance and color of the skin become similar to that of a toad	[Cha.Sa.Chikitsa Sthana 16/35]
Shwasa	Matta-rishabhaevashwasa in mahashwasa. In this condition, the	[Cha.Sa.Chikitsa

	breathing is accompanied by loud sound comparable to an intoxicated bull.	Sthana 17/46]
Purvarupa of kasa	Shookapoorna gala (thorns in throat). There is an intense itching sensation and pain in the throat as if there is presence of thorns in the throat.	[Cha.Sa.Chikitsa Sthana 18/5]
Kshatajakaasa	Paravataevakujana- the sound of cough similar to the humming of pigeon.	[Cha.Sa.Chikitsa Sthana 18/23]
Atisara	Appearance of stools in tridoshaja atisaara is like yakritkhandopama (similar to liver), mechakabham (like ashes)	[Cha.Sa. Chikitsa Sthana.19/9]
Agni visarpa	The vata-pitta prominence is observed in the pathology of visarpa where there is intense burning and pain similar to agni (fire)	[Cha.Sa.Chikitsa Sthana.19/35]
Trishna	Tapta-siktasutoya-description of the use of cold water to obtain relief in madyajatrishna as there is a reduction in the hotness of sand by pouring of the cold water	[Cha.Sa.Chikitsa Sthana.22/19]
Madatyaya	Tritiya madalakshana- bhagnadaruivanishkriya. This depicts the condition of a patient in the third stage of madatyaya where one appears to be comatose or immobile like a degraded piece of wood.	[Cha.Sa.Chikitsa Sthana.24/48]
Mutra-ashmari	Kadampushpakriti (similar to the kadambapushpa) – The renal calculi appears identical to the flower of Neolamarckia cadamba (kadamba)	[Cha.Sa. Chikitsa Sthana.26/37]
Vatarakta	Agnimaruta-tulyasya - pathology of vatarakta is considered to be identical to the combination of agni and vata	[Cha. Sa.Chikitsa Sthana.29/3]
Apatantraka	Kapota evakujeta - The sound or voice of the person becomes similar to a dove pigeon	[Cha.Sa.Siddhi Sthana.9/14]

It is useful in denoting the appearance, characteristics and morphology of the dosha involved with respect to its color (varna), smell (gandha), exudate (strava), type of pain (vedana) etc. and helps to differentiate between two types of same disease. For instance, the differential diagnosis of kushtha is largely based in the analogical aspects mentioned in the texts. For instance, the udumbarakushtha is similar to the fruit of fig, sidhmakushtha resembles the flower of lagenaria siceraria (alabu) while the kakanakakushtha resembles the abrus seeds. [Cha.Sa.Chikitsa Sthana.7/13-20]

The diagnosis of kshudrakushtha is largely based on the upamana pramana as these are compared with anecdotes like matsya-shakalopama (resembling the scales of a fish) like in eka-kushta, hasticharmavata (like the skin of an elephant) twacha in charma-kushta. [Cha.Sa.Chikitsa Sthana.7/21]

It is also helpful in determining the manifestation of disease pathology in the body like seed-soil theory i.e. the vitiation of a particular dosha in particular dhatu [Cha.Sa.Chikitsa Sthana.3/68], involvement of rasa dhatu and vyana-vayu in every disease manifestation. [Cha.Sa.Chikitsa Sthana.15/37]

Importance in treatment

Upamana pramana is applied to elaborate the effect of interventions on the physiological components like dosha, dhatu, srotas, mala, oja etc. that are difficult to assess. The similes in the form of upamana pramana help decipher the mode of action and consequences of the treatment procedures. Hence to elaborate on the mechanism and importance of digestive fire (agni) in the process of oleation (snehana), the example of a piece of cloth is illustrated.[Cha.Sa.Sutra Sthana.13/96] Similar to elaborating the oleation (snehana) and sudation (swedana) process, the example of softening of dried timber/wood after application of oleation and sudation is explained.[Cha.Sa.Sutra Sthana.14/5] The contribution of therapeutic enema (basti) in relieving the symptoms spread in the entire body is compared with the rising of the sun.[Cha.Sa.Siddhi Sthana 7/65] Similarly, the role of a physician in disease management is compared with different examples like a cook in cooking, ruler of the state etc.[Cha.Sa.Sutra Sthana 9/11,12] The selection of drugs is also based on the herb's inherent ability, for example, the use of plumbago (chitraka) in enhancing digestive abilities.

Similarly, the contra-indications in treatment are also expressed by giving examples like avoidance of purificaton (shodhana) in sama-avastha (condition associated with ama) of dosha. [Cha.Sa.Chikitsa Sthana.3/148] Upamana pramana highlights the benefits of consumption of certain dietary and drug interventions like buttermilk (takra) in diseases like piles (arsha). Buttermilk (takra) when spilled over the soil, arrests the growth of sapling, making it capable to heal the haemorrhoids in anal region completely.[Cha.Sa.Chikitsa Sthana.14/85] The mechanism of oleation (snehana) and sudation (swedana) in hiccups and dyspnea (hikka-shwasa) is also explained by quoting the example of melting ice.[Cha.Sa.Chikitsa Sthana.17/72]

Similarly, the action of application of torniquet (arishtabandhana) is compared to that of a bridge to arrest the spread of poison (visha) in the body.[Cha.Sa.Chikitsa Sthana.23/44] To indicate the use of milk (ksheera) in pacifying the poison (visha), the effect of rain on hot surroundings is compared.[Cha.Sa.Chikitsa Sthana.24/117] The importance of specific diet pattern (sansarjana

karma) in the ignition of digestive power (agni) after purification (shodhana) process is explained with the example of fire (agni) and figments (trina).[Cha.Sa.Siddhi Sthana.1/12]

Importance in teaching and learning methodology

Sushruta samhita describes various practical approaches towards learning and teaching surgical skills. The description of simulation and demonstration in training (yogyavidhi) [Su.Sa.Sutra Sthana.9/5][1] to develop proficiency in surgical skills. The appearance and design of various surgical instruments (yantra and shashtra) is also based on the upamana pramana for understanding their applications.[Su.Sa.Sutra Sthana.7][1], [Su.Sa. Sutra Sthana 8][1] Contemporary medical science accepts the application of analogy for the processes of learning, naming, remembering diseases.[10] The research has proved the relation between analogical knowledge and cognition along with different areas of brain.[11] Analogies contribute to meaningful learning and the visualisation of the facts. It is a tool for pedagogical learning for the application of knowledge.[12]

Utility in contemporary science

The medical science incorporated the principles of analogy to describe the different signs and symptoms of diseases rightfully. Many standard textbooks contain metaphors to explain the context. These are essential for medical teaching, clinical practice, memory-based learning; hence are vital for learning medical science. The simile of grapes is used to describe the different conditions like a hydatiform mole, tumors, active pulmonary tuberculosis appearance. The nomenclature of different diseases according to the similarities in appearance like snowstorm cataract, honeycomb liver, cobblestone appearance of body surfaces are examples of analogies. Many pathognomonic signs like strawberry tongue in Kawasaki disease, rose spots in typhoid, swan neck like deformity in rheumatoid arthritis, stepladder fever in malaria, moon facies in Cushing's disease, rice water like stools in cholera help in the precise diagnosis of the pathology.

Philosophical and other examples in Charak Samhita

Some references to different upamana mentioned in the Charaka Samhita are as below.

Table 3. Examples of upamana pramana

Entity / Phenomena	Upama given	Rationale	Reference
Mind (sattva), soul (atma), body (shareera)	tripod (tridanda)	Substratum for the existence of the being	[Cha.Sa.Sutra Sthana 1/46]
Medicine (aushadha)	nectar (amrita), poison (visha), fire (agni), lightening (ashani)	If properly prescribed, the medicine can act as elixir, while improper administration can be fatal as poison. Aushadha is also compared with fire and lightning as utmost care should be taken during its administration.	[Cha.Sa.Sutra Sthana 1/124, 126]
Benefits of collyrium (anjana)	moon in clear sky (nabhasi-induvata)	Administration of anjana makes the vision and eye like a moon in a clear sky.	[Cha.Sa.Sutra Sthana 5/19]
Importance of healthy regimen (swasthavritta) and daily regimen (dinacharya vidhi)	King of city, charioteer of chariot	The wise should take care of his/her body like an administrator takes care of their province.	[Cha.Sa.Sutra Sthana 5/103]
Harmful effects of excessive exercise (ati-vyayam)	Lion trying to drag an elephant	The body will perish with stress caused by excessive exercise, just as a lion in a powerful position perishes while trying to drag an elephant with higher strength.	[Cha.Sa.Sutra Sthana 7/35]
Importance of physician (bhishak)	Physician has been given the upama like cook, ruler, potter	As tasks like cooking food, ruling over a nation, preparing a pot are impossible without	[Cha.Sa.Sutra Sthana 9/11, 13]

		the cook, ruler, and potter despite all the tools. Physician (bhishak) is the most important entity in carrying out any task	
Importance of scientific study and logic (shastra dnyana and yukti)	Shastra is compared with Illuminating light, while the yukti or logic of the physician is compared with vision	Scientific scriptures provide light for illumination, and one's intellect is like eyes. The physician who uses both is capable of successful treatment.	[Cha.Sa.Sutra Sthana 9/24]
Description of anumana pramana	fire and smoke (dhooma and vanhi), sexual act and pregnancy (garbha and maithuna)	The examples of smoke (dhooma) and fire (vanhi) are used to describe the inferential knowledge regarding commonly occurring phenomena, while product of conception (garbha) and sexual act (maithuna) denote the inference regarding the happening of the past.	[Cha.Sa.Sutra Sthana 11/22]
Description of yukti pramana	process of farming (sasya-sambhava), conception (garbha-sambhava)	The examples of a process of farming and conception as these are multifactorial phenomena.	[Cha.Sa.Sutra Sthana 11/23-24]
Trayopastambha-Aahara, Nidra ,Bramhacharya	Three pillars of sustenance	The aahara i.e., food, nidra or sleep and bramhacharya i.e. proper sexual conduct, are considered as pillars of smooth functioning of life.	[Cha.Sa.Sutra Sthana 11/35]
Description of Vata	Various synonyms like yama ,	The synonyms are	[Cha.Sa.Sutra

dosha	prajapati	exemplarily explaining the qualities and functions of vata dosha.	Sthana 12/9]
Importance of purification (shodhana)	Dosha are compared with plants (druma)	To underline the importance and effect of shodhana therapy in which the vitiated tridosha are compared with a tree whose roots need to uprooted to weaken the tree.	[Cha.Sa.Sutra Sthana 16/21]
Description of ojas (essence of all body tissues)	colour similar to ghee (sarpivarna), smells like rice flakes (laja gandhi), honey (madhu)	The ghee (sarpi) and rice flakes (laja) help to denote the characteristics of ojas while the simile of honey (madhu) is representative of rasa or chemical composition of the same.	[Cha.Sa.Sutra Sthana 17/75]
Description of symptoms of conditions like shotha (edema)	pricking pain (suchibhirevatudyate), feeling similar to crawling of ants over the body (pippilikasancharanavatpeeda)	Different examples are given to assess the nature and intensity of pain.	[Cha.Sa.Sutra Sthana 18/7]
Manifestation of endogenous diseases primarily due to vitiation of tridosha (Nija vyadhi)	bird (Shakuni)	Here there is mention of a bird to denote the preponderance of dosha vitiation in any disease depicting the fact that bird cannot infringe upon its own shadow.	[Cha.Sa.Sutra Sthana 19/3]
Qualities of blood (rakta)	gold (suvarna), indragopa (insect), lotus (padma), abrus seeds (gunja)	These similes are mentioned in order to denote the qualities of pure blood.	[Cha.Sa.Sutra Sthana 24/22]
Description of	snakes who are hungry for	One must avoid getting	[Cha.Sa.Sutra

quacks	prey (sarpatepeetamaruta)	treated by quacks as this can be as dangerous as dealing with snakes hungry for prey.	Sthana 29/12]
Description of heart	central beam supporting the framework of roof (gopanasiagarakarnika)	The heart houses the important entities in the body just as the central beam supports the entire framework of the roof and house.	[Cha.Sa.Sutra Sthana 30/5]
Intake of ghee (ghritapana) in chronic fever (jeerna-jwara)	sprinkling water over burning house (veshmavaarisechana)	Just as water is sprinkled over the burning house to calm the fire, clarified butter (ghrita) is administered in chronic fever (jeerna jwara) to pacify the condition.	Cha.Sa.Nidana Sthana 3/37]
Appearance of urine (mutra) in prameha	gravel like urine (sikatameha), thread like urine (alalameha)	The pathology of abnormal urination (mutra pravritti) is described with the help of different upamana and illustrations depicting the appearance of urine (mutra).	[Cha.Sa.Nidana Sthana 4]
Aetiological factors of prameha	attraction of birds towards their nests (needa-drumaandaja)	The aetiological factors like lethargy (manda-utsaha), unctuous diet (atisnigdhaaahara) are responsible for the attraction of prameha in an individual just as the birds are attracted to their nests.	[Cha.Sa.Nidana Sthana 4/50]
Types of apatarpana chikitsa	Langhana is compared with a small quantity of water exposed to the sun and wind	Different upamana pertaining to pool, lake, and water are compared	[Cha.Sa.Vimana Sthana 3/45]

	dries up. Langhana-pachana with sink that is acted upon by the sun and wind on the one hand, and by a sprinkling of dust and ashes on the other side. Elimination of dosha (doshavasechana) with overflowing of a pool	with levels of severity of the dosha and their management with varying kinds of modalities like langhana, langhana-pachana and doshavasechana.	
Nomenclature of krimi (parsites/micro organisms)	similar to fig (Audumbaraakriti), similar to worm (gandupadakriti), similar to threads of wool (urnanshusankasha)	The nomenclature of krimi is based on the similarity of appearance of different krimi in the body.	[Cha.Sa.Vimana Sthana 7]
Characteristic feature of person with optimal semen/reproductive fluid (shukrasaarata)	milky white eyes (Ksheera-poornaivalochana)	The milky white eyes is used to describe the appearance of eyes in shukrasaara individual	[Cha.Sa.Vimana Sthana 8/109]
Importance of optimal body tissues (dhatu-saarata)	small ants carrying a big load (Pippilikabharaharanavat)	The importance of dhaatusarata in determining strength (bala) in an individual is denoted by the term pippilikabharaharanavat siddhi. The capability of a person depends on the quality of dhatu in person irrespective of body type.	[Cha.Sa.Vimana Sthana 8/115]
Importance of logical reasoning (yukti)	seed for farm (beejamevakarshakasya)	Yukti is the guiding force for any physician in the process of decision-making like seeds for a farmer.	[Cha.Sa.Vimana Sthana 8/149]
Importance of soul (atma)	potter (kumbhakara), mason (grihakara)	The atma is the driving force of body and is as important as the kumbhakara or potter in	[Cha.Sa.Sharira Sthana 1/43-44] [Cha.Sa.Sharira Sthana 1/74]

		preparation of pot and grihakara or mason in building a house.	
Desires (trishna) as cause for miseries (dukkhakarana)	silk (koshakara)	Just as a silkworm provides for itself suicidal threads, the individual who is unaware of trishna gets succumbs to miseries (dukkha).	[Cha.Sa.Sharira Sthana 1/93]
Embryo (garbha) as a resultant of multiple factors	house (kutagara), chariot	Just as many objects are involved in the construction of houses and chariots, embryo (garbha) is the resultant of multiple factors	[Cha.Sa.Sharira Sthana 3/14]
Description of mental constitution (manas prakriti)	Brahma sattva, sarpasattva, vanaspatya sattva	The determinants of manas prakriti are based on the examples like snake (sarpa) , tree (vanaspatya)	[Cha.Sa.Sharira Sthana 4/36]
Appearance of organs of foetus (garbhaavayaya)	palms of new-born similar to shape of tortoise (kurmakara hasta)	In a healthy infant, the palms of the new-born should be shaped like a turtle.	[Cha.Sa.Sharira Sthana 8/51]
Description of voice (swara)	swan (hansa), stark (krauncha)	The normal and abnormal swara or voices are compared with different animals and objects to understand the quality and intensity of the voice.	[Cha.Sa.Indriya Sthana 1/14]
Description of abnormal odour (vikrita gandha)	sandal (chandana), dead body (kunapa)	The normal and abnormal odour (gandha) from the body are expressed using	[Cha.Sa.Indriya Sthana 2/8-16]

		different examples like sandal	
Appearance of lips (austha)	ripened Syzgium cumini (jambu) fruit (pakvajambunibha)	The appearance of the lips in a person about to die are similar to ripened jambu fruit	[Cha.Sa.Indriya Sthana 3/6]
Fatal signs (arishta) related to eye (chakshurendriya)	solid sky (ghanibhoota akasha)	The different upama are used to denote the appearance of the sky for the person about to die	[Cha.Sa.Indriya Sthana 4/7-9]
Description of complexion and lustre (chaya)	similar to gem like cats eye (Vaiduryadrishti)	The shadow (chhaya) due to jala mahabhuta is similar to gem like cats eye.	[Cha.Sa.Indriya Sthana 7/12]
Signs of impending death (sadyamarana)	teeth as if covered with mud (kardamavatadanta)	Different signs like the appearance of teeth as if they are covered with mud depict the impending death.	[Cha.Sa.Indriya Sthana 10/29]
Importance of aphrodisiac treatment (vajikarana)	comparable to horse (vajivata), huge tree (mahachaitya)	Vajikarana renders the individual with the vitality and sexual drive similar to that of a horse. It is also responsible for the fertility or the advancement of pedigree like the growth of trees (mahachaitya).	[Cha.Sa.Chikitsa Sthana 1/9-12]
Benefits of consuming different rejuvenating (rasayana) herbs	glow like a sun (aditya-samadyuti), stability of the body (dharanidharasaara), nectar (amrita)	Consumption of different rasayana combinations can render different advantages like glow like the Sun, stability of body etc. are expressed through various examples mentioned before.	[Cha.Sa.Chikitsa Sthana 1]

Types of black bitumen (shilajatu)	like the feathers of peacock (mayurpicchasama), similar to hibiscus (japapushpasama), similar to cows urine (gomutragandhi)	The appearance of different types of shilajatu is compared with objects like mayur-piccha or peacock feathers etc.	[Cha.Sa.Chikitsa Sthana 1/3/56-61]
Description of semen (shukra) and its expression	Sugar in ikshu, ghee in curd, taila in sesame seeds.	Different examples like the juice exist in the sugarcane, ghee in the curd, and oil exists in the sesame seed, semen pervades the entire body, which has the sensation of touch. This shukra is expressed at the right time, just as the ghee melts when in proximity to the heat.	[Cha.Sa.Chikitsa Sthana 2/46], [Cha.Sa.Chikitsa Sthana 15/35]
Description of process of digestion according to Ayurveda	cooking of rice in an utensil (odana pachana)	The process of digestion of food is explained by comparing it with the cooking of rice in utensils where the heat is applied to the base of utensil. Here the heat is compared with digestive fire (jatharagni) while the utensil is amashaya or digestive tract.	[[Cha.Sa.Chikitsa Sthana 15/8]
Explanation of functioning of heart (hridaya)	lotus (pundarika)	The body's functioning is more smooth during the day, just as the lotus blossoms with the rise of the day.	[Cha.Sa.Chikitsa Sthana 15/238]
Description of alcohol (madya)	Madya is compared with anna (food)	Madya is compared with anna as both of these can have beneficial and	[Cha.Sa.Chikitsa Sthana 24/59]

		harmful effects if not used rationally.	
Description of menstrual blood (artava)	abrus seeds (Gunjaphalasavarna), lotus (padmasama), insect (indragopasama)	The colour , quality of pure menstrual blood (shuddhaartava) are described by the nouns like abrus (gunja) etc.	[Cha.Sa.Chikitsa Sthana 30/226]
Description of medium consistency (madhyama paaka)	like a soft porridge (Sanyavaivaniryava)	The madhyama paaka of oil (taila) is compared with porridge like consistency	[Cha.Sa.Kalpa Sthana 12/103]
Importance of therapeutic enema (niruha basti)	cloth (pata)	Niruha eliminates the disease from the body just as a cloth removes the dirt when mixed with water	[Cha.Sa.Siddhi Sthana 7/65]
Description of three vital points (marma)	axles of wheel (Nabyamivaamara), rays of sun (suryamivagabhastaya) , udadhievaapaganam	The three marma namely heart (hridaya), head (shira) and bladder region (basti) function like axles of wheel, rays of sun, rivers flowing into the ocean, respectively.	[Cha.Sa.Siddhi Sthana 9/5]
Importance of taking care of patient post the purifiction (shodhana) therapy	unfertilized egg (tarunaanda), utensil filled with oil (poornataila-patra), shepherd (gopala)	After the administration of shodhana therapy in patients, proper care of patient should be taken as the condition of the patient is as delicate as a raw, unfertilized egg, utensil filled with oil or the way a shepherd takes care of one's herd.	[Cha.Sa.Siddhi Sthana 12/5]
Description of tantrayukti	lotus lake (Ambujavana), lightened lamp	The tantrayukti throws light upon the subject and helps to understand its hidden meaning, just	[Cha.Sa. Siddhi Sthana 12/46]

		as the lotus blossoms on the rise of the sun or a room becomes lit after the lamp is lightened.	

Current researches

1. Dadu V, Purohit H. have explained the principles of epistemology in the perspective of Nyaya Philosophy and have compared upamana pramana with analogy explaining the application of the same in research methodology.[13]
2. Sharma S.et.al. explained the scope of upamana pramana in the diagnosis of different vyadhi according to Ayurveda with the help of pictorial depictions.[14]
3. Bhuvaneshwari C et. al. explored the utility of upamana in all the domains of physiology, diagnosis, treatment, philosophy of Ayurveda.[15]
4. Anjana Priya has depicted the importance of upamanapramana as an evidence based method to understand the Ayurveda using a bird's eye view methodology.[16]
5. Jaggi S. et. al. have elaborated on the role of upamana pramana in the identification of plants, determining the diagnosis and prognosis of diseases.[17]
6. Trupti Arun Jadhav and Rajendra K. Patil mentioned the inputs of upamanapramana in understanding Sandhi Shareera.[18]
7. Bhaskar Jha elaborated the philosophical aspects of upamana pramana in obtaining appropriate knowledge considering the nyaya philosophy.[19]
8. Phule S. et. al. explains the importance of upamana pramana in understanding physiological concepts like prakriti and saarata according to Ayurveda.[20]
9. Sharma C, Bharadwaj S. explained the role of upamanapramana as an essential component in comprehension, application, and synthesis of knowledge of the learning of Ayurveda sciences and its application in the postgraduate competitive examinations in the letter to the editor.[21]
10. Jha C et. al. explored the utility of upamana in the development of gadgets concerning biological mechanisms using the basis of upamana pramana.[22]
11. Wargheda M. et. al. focus on the importance of upamana pramana in the diagnosis of diseases.[23]
12. Bhalerao S .et. al. explained the role of upamanapramana as imaginative creativity to analogize for evidence and its application in the pedagogy in the letter to the editor.[24]

13. Yogita C. explained the importance of upamana pramana as a method of examination of diseases.[25]
14. Vasant Patil, Sapra U. explained the upamana pramana as a tool in the diagnosis of different disorders.[26]
15. V. J. Thaker explains upamana as the method of comparison and contrast while diagnosing a disease.[27]
16. In an archive from Stanford Encyclopaedia of philosophy archives entitled upamana is elaborated as presupposing knowledge of relevant similarity, which can be a matter of perception considering the Indian philosophy.[28]
17. Navya K et. al. beautifully elaborated the application of upamana pramana in understanding the Sushrut Samhita and principles of diagnosis and treatment explained in the same considering all the upamana.[29]
18. Shivappa et. al. underlined the importance of upamana pramana in clinical practice and diagnosis.[30]
19. Anoop A et. al. explained the applicability of upamana pramana in several aspects in the current scenario like philosophy, anatomy, diagnosis, treatment using pictorial depiction in the article entitled.[31]
20. Shubada V. has explained the importance of upamana in different subjects of ayurveda.[32]
21. In a review article, Sanyal K. has explained the nuances of upamana pramana concerning philosophical knowledge and its application in understanding Indian epistemology.[33]
22. M.V. Vinodkumar, A.K. Anoop describe upamana as a crucial epistemological method and analogical processes and symbolic expressions in explaining theories.[34]
23. Padashetti S. has described different examples of upamana pramana mentioned in the Charak Samhita.[35]
24. Tejaswini J. and Vinay Hiremath described the upamana pramana in understanding the arishta lakshana and indriya sthana of charak Samhita.[36]
25. Rashmi T. et. al. mentioned different analogies associated with the shape and appearance of body ulcers.[37]
26. Vyas K et. al. mentioned upamana pramana as an important vaadamarga in research and treatment.[38]
27. Mahajan N et. al. describe the application of upamana pramana in assessing the abnormalities in urine (mutra-vikriti) in disorders like prameha.[39]

28. Patwardhan B et. al. termed upamana pramana as associative conjunctures, similes and analogies and its application in producing the vaidya.[40]
29. Vaidya A. described pramana including upamana pramana, as a mode of evidence for Ayurveda research and evidence generation.[41]
30. Rashmi T. et. al. described the analogical understanding of wound and its exudates and their application in diagnosis and treatment.[42]
31. Gede A et. al. mentioned upamana pramana as an important part in developing original science content and pedagogy context of four types of pramana.[43]
32. Jaggi S. et. al. in the article has focused on the literary aspects and utility of upamana pramana.[44]

Related chapters

Tistraishaniya Adhyaya, Rogabhishagjitiya Vimana, Trividha Roga Vishesha Vijnaniya Vimana Adhyaya, Pramana, Pratyaksha pramana, Anumana pramana,Aaptopadesha pramana, Yukti pramana, Apamarga Tanduliya Adhyaya, Concepts and Contemporary Practices, Naveganadharaniya Adhyaya

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